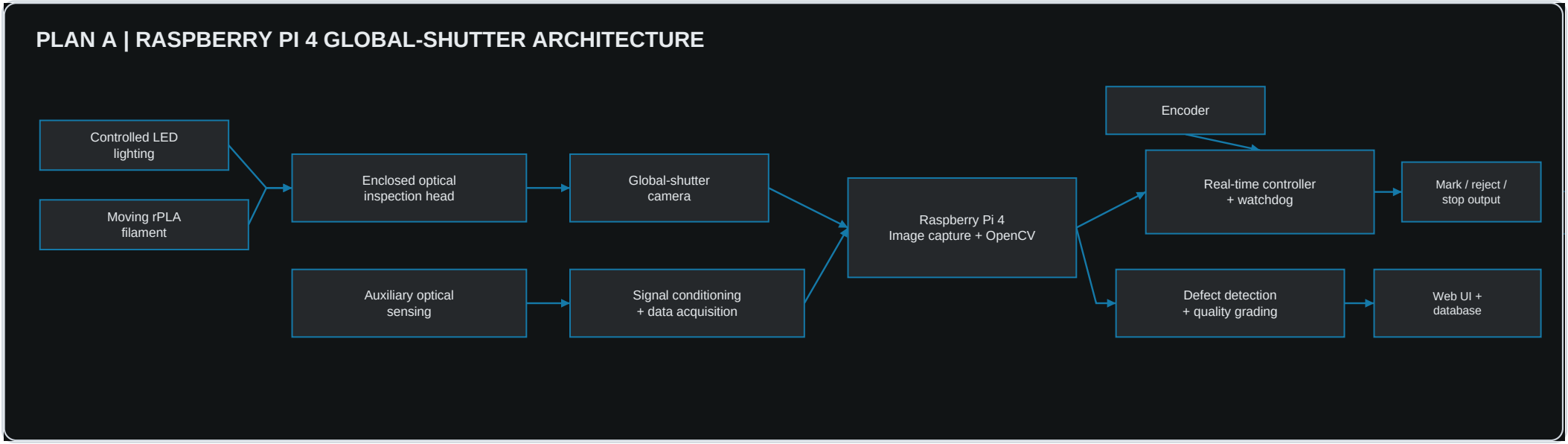
Plan A - Recommended Final Prototype

rPLA filament surface quality monitoring using Raspberry Pi 4

DECISION REQUEST: APPROVE ESSENTIAL ITEMS FIRST



Reading guide: controlled imaging and auxiliary sensing feed Raspberry Pi processing; real-time control supports position tracking and safe action.

Why Plan A is needed

- Global-shutter imaging reduces motion distortion on moving filament.
- Direct camera connection avoids Wi-Fi delay and video-compression artefacts.
- A selected lens and controlled lighting enable repeatable calibration.
- Encoder data links every detected defect to a physical filament position.

Staged funding request

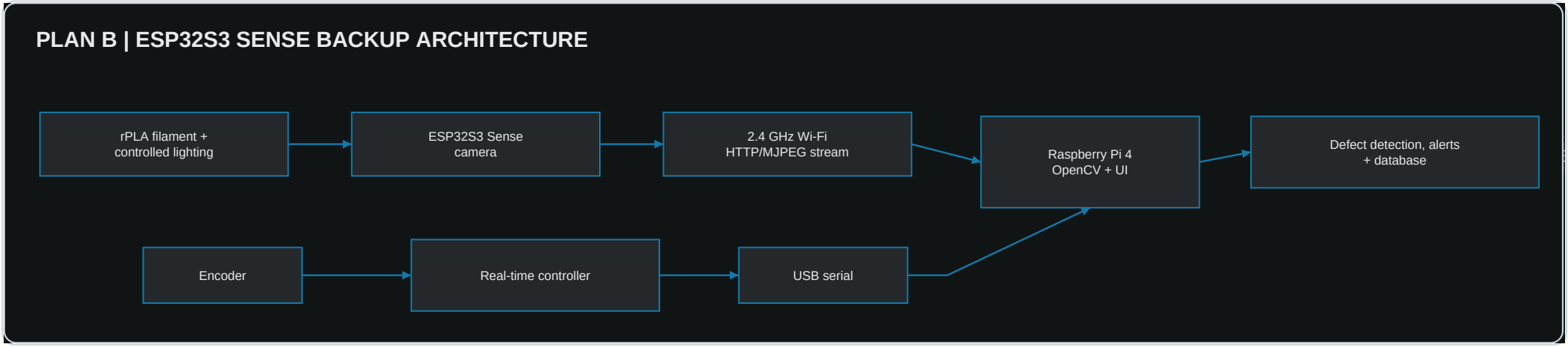
Imaging: camera and selected lens	A$136.15
Measurement: optical encoder	A$41.00-A$43.00
Controlled lighting	A$30.00-A$60.00
Safe auxiliary power	A$30.00-A$50.00
Essential request	**A$237.15-A$289.15**
Full request including mechanical and optional items:	**A$277.15-A$384.15**

Prices are planning estimates and will be rechecked before purchase. GST and shipping are excluded unless stated by the supplier.

Plan B - Cost-Controlled Backup

Supplied ESP32S3 Sense camera supports early development and the Week 6 demonstration

NO ADDITIONAL CAMERA PURCHASE FOR THE BACKUP DEMO



Reading guide: the ESP32S3 camera sends compressed video over 2.4 GHz Wi-Fi; Raspberry Pi 4 performs detection, display and data logging.

Why retain Plan B

- Uses the supplied camera for immediate development.
- Protects the Week 6 milestone if purchased parts are delayed.
- Reuses the same Raspberry Pi processing, interface and database.

Why it is not the final solution

- Rolling shutter can distort a moving filament.
- MJPEG compression and Wi-Fi add artefacts, delay and connection risk.
- Small-defect validation requires more controlled and repeatable optics.

Decision requested

Could the unit support the essential Plan A purchase? If the full amount is unavailable, we request a staged approval and guidance on the available budget cap.

Plan B remains available as a documented fallback. If funding is not approved, final speed and minimum-defect claims will be formally reduced.